Mahdi Farmahini Farahani **(Arad)** **Ph.D. Student**
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EDUCATION

The Chinese University of Hong Kong (CUHK) Hong Kong SAR
Doctor of Philosophy, Geophysics (Seismology) *2026 – 2030*

Kharazmi University, Faculty of Earth Sciences Tehran, Iran
Bachelor of Science, Geology *2021 – 2025*
- Chairperson, Geological Society of Kharazmi University (June 2024 – June 2025)
- Vice President of Media, Geological Society of Kharazmi University (November 2023 – June 2024)

University of Lille, International Academy Lille Hdf Lille, France
Pre-Ph.D. Water Resource Management Summer School (5 ECTS) *Summer 2024*
- Presented a numerical groundwater modeling study on water abstraction impacts in the Weyib Watershed, Ethiopia. Under Supervision of the Prof. Arnaud Gauthier.

TEACHING EXPERIENCE

CUHK Hong Kong SAR
Hydrogeology Teaching Assistant *August 2026 – Present*
- (EESC3320), under supervision of Prof. Yen Joe Tan

Space Research and Technology Institute (SRTI) Primorsko, Bulgaria
Lecturer *August 2026 – September 2026*
- EPEC ML WG Summer School 2026, under supervision of Prof. Lachezar Filchev

Iranian Young Scholars Club (IrYSC) Tehran, Iran
Earth Science Olympiad Teacher *April 2021 – August 2026*

Kharazmi University Tehran, Iran
Programming Teaching Assistant *April 2025 – June 2025*
- Programming Course, under supervision of Prof. Morteza Asemani

Kharazmi University Tehran, Iran
Geostatistics Teaching Assistant *February 2025 – June 2025*
- Geostatistics Course, under supervision of Prof. Ehsan Pegah

Young Scholars Club Tehran, Iran
Instructor *July 2024*
- Taught "Applied Data Science in Geology" for the Iran National IESO 2024 Team

Kharazmi University Tehran, Iran
Instructor *April 2024 – June 2024*
- Organized and instructed the Advanced Python and Machine Learning Workshop

National Organization for Development of Exceptional Talents Tehran, Iran
Teacher *May 2021 – September 2021*
- Taught ITEO Marketing at Allameh Helli (Branches 5 & 8) and Farzanegan High Schools

RESEARCH EXPERIENCE

FEZtool Tehran, Iran
Co-Founder & Research and Development Specialist *August 2023 – Present*
- Led Earth science research using advanced algorithms and geospatial analysis.

Center for Complex Networks, SBU Partner Tehran, Iran
Research Assistant *November 2025 – June 2026*
- Worked on fashion trend forecasting project under supervision of Prof. Reza jafari.

International Astronomical Search Collaboration, NASA Partner Remote
Asteroid Search Team Member *July 2025 – September 2025*
- Analyzed Pan-STARRS data, contributing to 3 preliminary asteroid discoveries.

Kharazmi University Tehran, Iran
Research Assistant *October 2024 – January 2025*
- Performed MASW to extract dispersion images under supervision of Prof. Mehdi Talkhablou.

COMMUNITY SERVICE

Zamin Pouya Scientific Journal Vol. 31 Tehran, Iran
Associate Editor & Reviewer *August 2025 – June 2026*

The 4th National Festival of Young Iranian Geologists Tehran, Iran
Education Committee Member *April 2025 – May 2025*

The 3rd National Festival of Young Iranian Geologists Tehran, Iran
Graphics Committee Member *May 2024 – June 2024*

Imam Khomeini International University Qazvin, Iran
Editorial Board Member *September 2023 – June 2024*

PUBLICATIONS

Journal Articles

1. **Farmahini Farahani, M.**, Elmi, P., Nedaee, M., Kianifaizabadi, M., Mirzaee, H., Moradi, P., Yazdanifar, M. Khosravani, F., Najafi, F., & Talkhablou, M. (2026). FEZrs: An Open-Source Python Package for Geospatial Multispectral Image Processing and Remote Sensing. ***Journal of Open Source Software (JOSS).***
2. Talkhablou, M., **Farmahini Farahani, M.**, & Mansouri, S. S. (2025). A Python-Based Framework for Land Cover Classification in Engineering Geology: A Comparative Assessment of SVM, K-Means, and Spectral Indices. ***Kharazmi Journal of Earth Sciences.***
3. **Farmahini Farahani, M.** (2024). Effects and social damage of fine dust. ***Ovan Scientific Journal***
4. **Farmahini Farahani, M.** (2024). Designing the green belt in the southwest of Zahedan. ***Ovan Scientific Journal***
5. Saravani, M. J., Kashef, S., **Farmahini Farahani, M.**, Kashefi, M., & Zohreh, M. (2023). Investigating the accuracy of hybrid models with wavelet transform in the forecast of watershed runoff. ***Journal of Water Management Modeling (JWMM)***

Conference Papers

1. **Farmahini Farahani, M.**, & Bagheri, M. (2026, November). Attention Guided Dual Dilated CNN for Enhanced Random Noise Suppression in Seismic Records. In ***22nd Iranian Geophysical Conference (IGC22)***, Tehran, Iran.
2. Khosravani, F., **Farmahini Farahani, M.**, & Bagheri, M. (2026, November). Fourier Neural Operators with pointwise Convolutional Neural Networks for Seismic data random noise attenuation. In ***22nd Iranian Geophysical Conference (IGC22)***, Tehran, Iran.
3. **Farmahini Farahani, M.** (2025, August). Machine Learning-Enhanced Seismic Event Detection and Noise Reduction for Planetary Seismology: Insights from Mars and Lunar Data. In ***Seismology Student Workshop (SSW 2025)***, GatherTown (Virtual).
4. Rezaei, K., & **Farmahini Farahani, M.** (2024, March). Visualization of magnetometric data with Python (case study: Segregation sedimentary units of Torud Playa). In ***The 1st Conference on Applied Geophysics in Mines***, Tehran, Iran.
5. Rezaei, K., & **Farmahini Farahani, M.** (2024, February). Comparing the performance of the k-means clustering method and spectral indices in characterizing land use and land cover using remote sensing with Python (case study: Landsat 8 multi-spectral images). In ***2nd International Conference for Iranian Geography and Earth Sciences Students***, Tehran, Iran.
6. **Farmahini Farahani, M.**, Rezaei, K., Karamollahi, S., & Moghadam, F. J. (2024, January). Application of PCA algorithm in remote sensing using Python programming language (case study: Landsat 9 multispectral images). In ***8th Symposium of Sedimentological Society of Iran***, Hormozgan, Iran.
7. Rezaei, K., & **Farmahini Farahani, M.** (2023, September). The importance of using Python programming language in Earth science. In ***26th Conference of Geological Society of Iran***, Urmia, Iran.

Books

1. **Farmahini Farahani, M.**, Najibifar, Y., & Akbarzadeh Afshari, A. (2025). *Earthquake Prediction Using Geophysical Data Analysis and Artificial Intelligence.* **Heritage Branch, Library and Archives Canada**
2. Rezaei, K., **Farmahini Farahani, M.**, Karimzadehasl, E. (Trans.). (2024). *Introduction to Python in Earth science data analysis: From descriptive statistics to machine learning.* **Tasir Publication.** (Original work published by Maurizio Petrelli)
3. Karimi Pashaki, A., **Farmahini Farahani, M.**, Soleimani, J., & Karimzadegan, Y. (In Farsi). (2023). *Environmental Risk Management.* **Simorgh Aseman Azargan Publication.**

PROJECTS

PlanetaryGeophysics | `github.com/aradfarahani/PlanetaryGeophysics` *2025 – Present*
- Built an open-source Python platform for analyzing and visualizing planetary geophysical data.

FEZrs (Open Source) | `pypi.org/project/fezrs` *2023 – Present*
- Advanced Python library for remote sensing, developed by FEZtool with over 100,000 downloads

Seismicity Analysis in Iran (2010–2025) | `iris.edu/hq/workshops/2025/06` *2025*
- performed Iran seismicity analysis for the *Seismology Skill-Building Workshop 2025* final project.

Geoelectricspy (Open Source) | `zenodo.org/records/10966815` *2023*
- Interactive 3D visualization tool for subsurface resistivity using geoelectrics survey data

INVITED TALKS

1. **Seismic Data Denoising in the Era of Neural Networks:** *5th National Festival of Young Iranian Geologists (Hosted by Mohaghegh Ardabili University, Kharazmi University, RIES, & UNESCO Chair);* 17 October 2026

2. **Walrus: A Cross-Domain Foundation Model for Continuum Dynamics:** *Applied ML Lab, Center for Complex Networks, Department of Physics, Shahid Beheshti University;* 13 December 2025
3. **Applied Artificial Intelligence in Earth Science:** *Faculty of Geosciences, Shahrood University of Technology & Faculty of Earth Sciences, Kharazmi University;* 11 November 2025
4. **Applications of Artificial Intelligence in Education and Earth Science Analysis:** *Educational Technology Expert Group, Karaj and Fardis County Department of Education, District 2;* 30 December 2024

LICENSES & CERTIFICATIONS

Applied Geotechnical Seismology, Kharazmi University *2026*
Applications of Non-Destructive Geophysical Methods in Engineering Geology Studies, Kharazmi University
Seismology Skill Building Workshop 2025, EarthScope Consortium *2025*
Investigating practical issues in the exploration of Sulphide Deposits using the IP-RS method, National Iranian Geophysical Society
Remote Pilot Certification (Multirotor), Civil Aviation Authority of Islamic Republic of Iran *2023*
Unmanned Aerial Vehicle Surveying, Maan Academy
GIS, Mapping, and Spatial Analysis Specialization, University of Toronto (Offered by Coursera)
CS50 Certificate, CS50 (Harvard University) *2021*

HONORS & AWARDS

Geology & Geophysics Photo Contest – *Student Winner, EEGS* 2026
CUHK's International Student PhD Fellowship – *Awardee, CUHK*
9th National Young Scientists Festival – *Merit Award Winner, Geology Category* 2025
Hydro 2024 Research-Oriented Summer School – *Full Scholarship, University of Lille* 2024
Research Visit Grant (2500 DKK) – *Awardee, University of Copenhagen* 2022
DrCT 2021 Global Computational Competition – *Gold Medalist* 2021
1st National Thinking & Entrepreneurship Olympiad (ITEO) – *Silver Medalist* 2020

SKILLS & INTERESTS

Programming Languages: Python (ObsPy, pyGIMLi, Astropy, OpenCV, TensorFlow, PyGMT, Xarray, Scikit-learn, kepler.gl, Leafmap, Folium, Rasterio, GeoPandas), Julia, Bash (Shell scripting), SQL (MySQL), GMT, GDAL; Familiar with R
Markup Languages: Markdown, LaTeX
Softwares: Astrometrica, ENVI, Seismic Analysis Code(SAC), SeisAn, QGIS, SeisImager, Blender, ilastik, Google Earth Pro, Refrapy, SAS.Planet, draw.io, Groundwater Vistas
Field Techniques: Drone Surveying, Geoelectrical Methods (Induced Polarization–Resistivity Sounding, IP-RS), Seismic Refraction Data Acquisition
Languages: Farsi (Persian): *Native*; English: *Fluent (C1)*; German: *Elementary (A2)*; Mandarin: *Beginner (Duolingo Score: 11)*
Research Interests: Signal/Image Processing, Geophysics (Seismology), Computational Science, Open-Source Software Development

PROFESSIONAL MEMBERSHIPS

National Iranian Geophysical Society Student member, 2025 – Present
Space Generation Advisory Council Member, 2025 – Present
Google Maps Platform Innovators Member, 2025 – Present
Geological Society of Iran Student member, 2021 – 2022